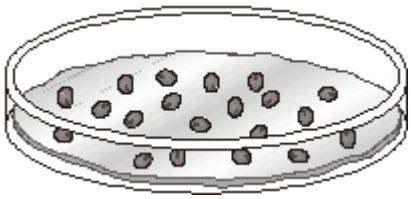
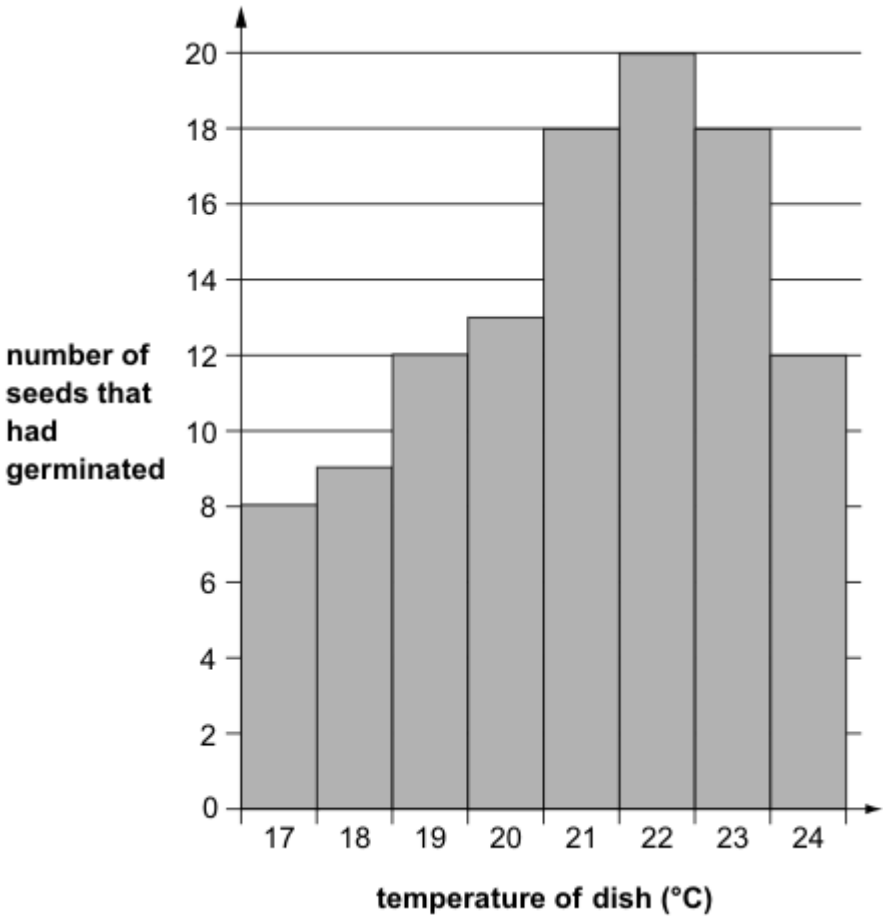
Q1. Abdul put cress seeds on wet filter paper in dishes.
He put 20 seeds in each dish.
Every day he added 5 cm^3 of water to each dish.
He kept each dish at a different temperature.



The bar chart below shows how many seeds had germinated after two days.



Use the bar chart to answer the following questions.

(a) (i) How many different temperatures did Abdul use?

.....

1 mark

(ii) What was the lowest temperature Abdul used?

..... °C

1 mark

(iii) How many seeds had germinated at 21°C?

.....

1 mark

(iv) Abdul said 23°C was better than 21°C for seeds to germinate.
Was he correct?
Tick the correct box.

yes ☐ no ☐

Use the bar chart to help you give a reason for your choice.

.....

.....

1 mark

(v) How does the bar chart show that 22°C is the best temperature for seeds to germinate?

.....

.....

1 mark

(b) Give **one** way Abdul made sure his investigation was a fair test.

.....

1 mark
maximum 6 marks

Q1.

(a) (i) • 8

1 (L3)

(ii) • 17 °*C*

1 (L3)

(iii) • 18

1 (L4)

(iv) • no ✓

if more than one box is ticked, award no mark
***both** the answer and the correct explanation are required for the mark*

any **one** from

- 18 seeds germinated at both temperatures
- the number that germinated was the same
  accept 'the results were the same'
  accept 'the bars are the same height'

1 (L4)

(v) any **one** from

- all the seeds germinated **or** grew
  *accept 'they all germinated **or** grew'*
- 20 seeds germinated **or** grew
- the most **or** more seeds germinated **or** grew
  accept 'the bar is higher'

1 (L3)

(b) any **one** from

- he used the same number of seeds
  accept 'the same amount of seeds'
  'number of seeds' is insufficient
- he counted them after two days **or** after the same time
  'he counted the seeds' is insufficient
- he used the same type of seeds **or** they were all cress seeds
- he used filter paper to grow all the seeds on
- he used 5 cm^3 of water each day
  accept 'the same amount of water'
  'he used the same sized dishes' is insufficient

1 (L4)

[6]